

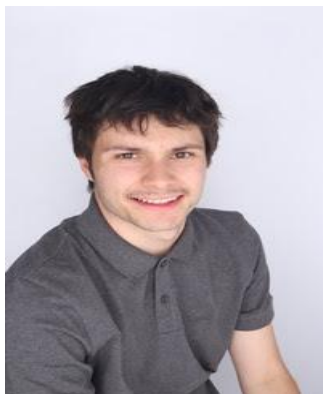
Urologen Inc.

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The Problem: CAUTIs And Current Catheter Design

- 15–25% of hospitalized patients receive urinary catheters
- Catheter-Associated UTIs (CAUTIs) account for 70–80% of hospital-acquired UTIs
- Backflow of urine contributes significantly to infection risk
- Many patients report discomfort, pain, and complications with current catheters
- Current designs lack reliable unidirectional flow mechanisms



Market Snapshot: The Growing Catheter Market

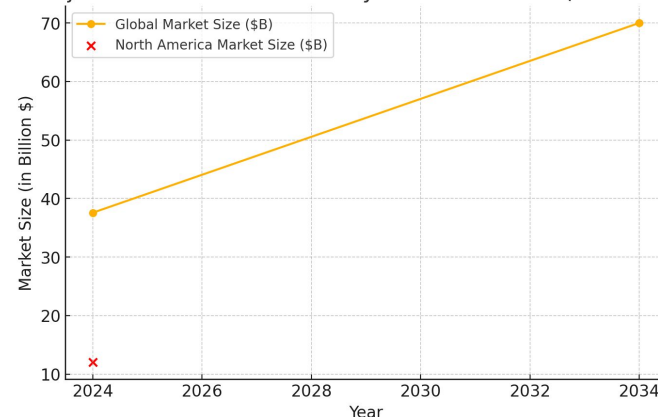
Market Size and Growth:

- ❑ The global urinary catheter market is projected to grow from \$37.58B in 2024 to over \$70B by 2034
- ❑ North America's market was evaluated at \$12.03B in 2024

Cost and Use:

- ❑ Catheters range from \$1 to \$10 per unit and are often reimbursed by insurance providers

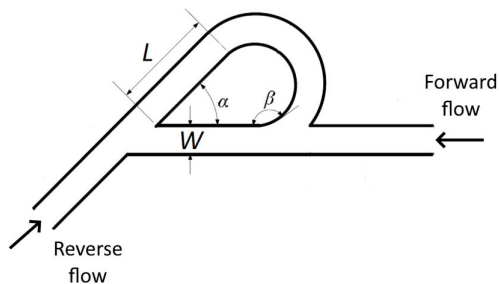
Projected Growth of the Urinary Catheter Market (2024-2034)



Product Opportunity And Competitive Landscape

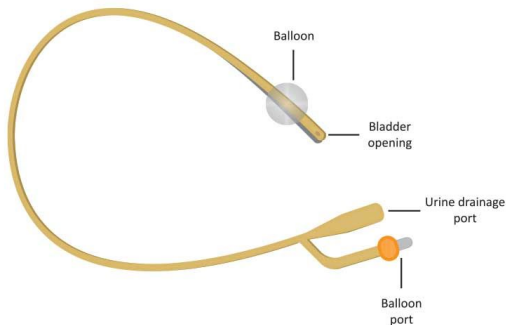
Our Innovation: Tesla Valve Catheter

A one-way check valve catheter leveraging Tesla flow dynamics to prevent urine backflow, minimizing CAUTIs without mechanical parts.



Competitive Products:

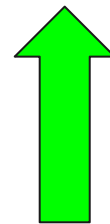
Current solutions include mechanical valves, antimicrobial coatings, and absorbent materials—often more invasive or prone to failure.



Differentiators:

Our catheter eliminates moving parts, improves reliability, and enhances patient comfort through an external valve placement.

Patient Outcomes





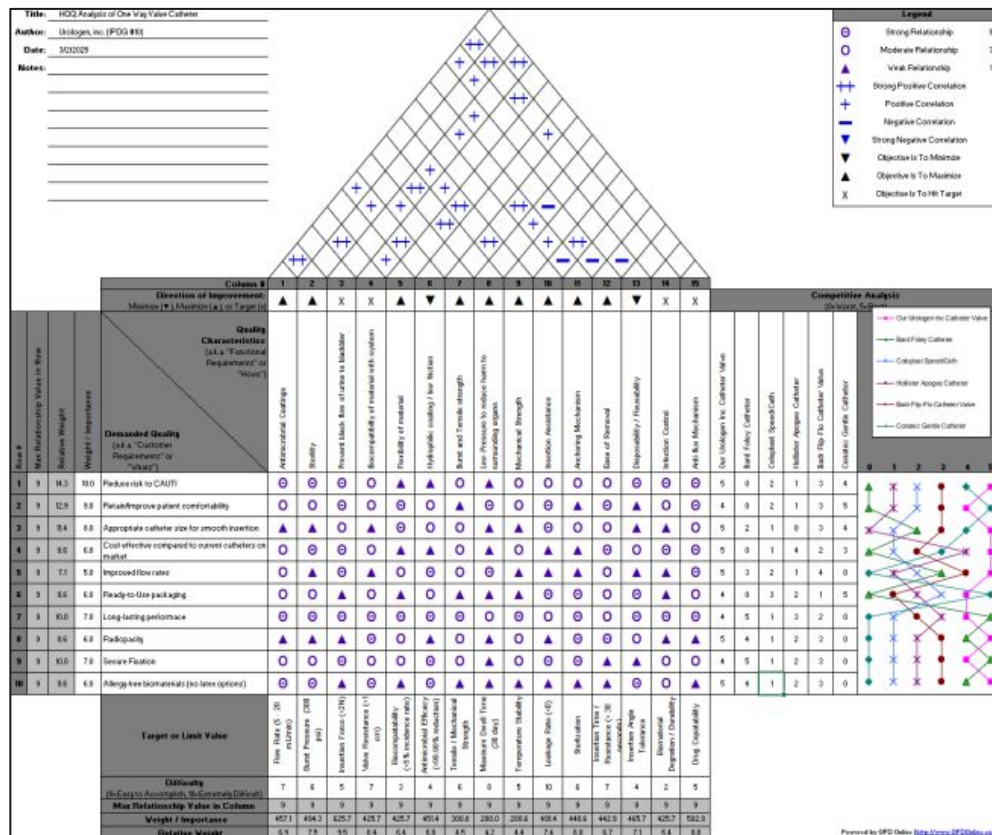
Voice Of The Customer

Improved Patient Needs:

- ❑ Longevity
- ❑ Comfort

Common Issues:

- ❑ Leaks from material degradation
- ❑ Mechanical failures
- ❑ Causing UTIs or irritation



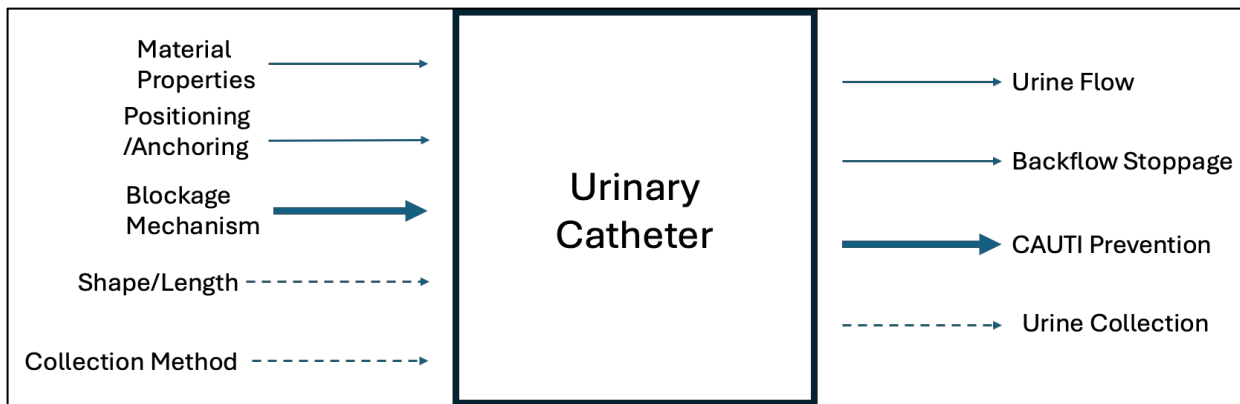
Technical Requirements

- ❑ Unidirectional flow
- ❑ Durable, biocompatible materials
- ❑ External valve placement for comfort and control
- ❑ Scalable design for hospital use
- ❑ Easy integration with existing catheter systems
- ❑ Minimized added cost per unit

Row #	Max Relationship Value in Row	Relative Weight	Weight / Importance	Quality Characteristics (a.k.a. "Functional Requirements" or "Hows") Demanded Quality (a.k.a. "Customer Requirements" or "Whats")
1	9	14.3	10.0	Reduce risk to CAUTI
2	9	12.9	9.0	Retain/Improve patient comfortability
3	9	11.4	8.0	Appropriate catheter size for smooth insertion
4	9	8.6	6.0	Cost effective compared to current catheters on market
5	9	7.1	5.0	Improved flow rates
6	9	8.6	6.0	Ready-to-Use packaging
7	9	10.0	7.0	Long-lasting performance
8	9	8.6	6.0	Radiopacity
9	9	10.0	7.0	Secure Fixation
10	9	8.6	6.0	Allergy-free biomaterials (no latex options)

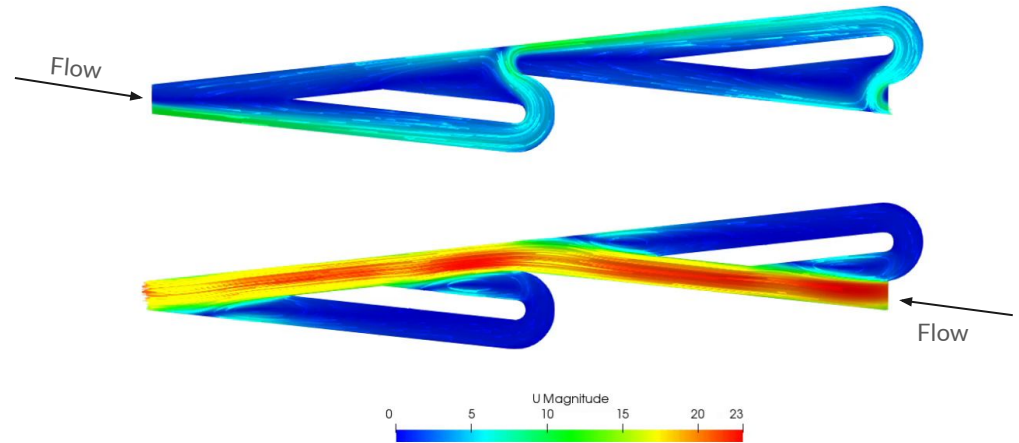
Concept Justification - Addressing Customer Needs

- ❑ Reduces CAUTI risk through backflow prevention
- ❑ Minimizes need for replacement and maintenance
- ❑ Allows hospital and home implementation
- ❑ External valve design → prevents further discomfort
- ❑ Compatible with existing systems → ease of adoption
- ❑ Addresses clinician concerns over insertion trauma



Technical Model Part 1- Testing The Concept

- ❑ Computational Fluid Dynamics (CFD) Model
- ❑ SOLIDWORKS 3D Flow Simulation
- ❑ Use fluid properties of urine as modeling fluid.
- ❑ Mathematical Justification:
 - ❑ Reynolds number calculation
 - ❑ $Re = vde/\lambda$
 - ❑ Confirm urine flow rates are within the normal range of 0.732-4.448 cc/sec



Path to Market

Regulatory Pathway:

Urologen Inc. plans to pursue a FDA 510(k) submission, Class II device

Development Timeline:

Prototype within 12 months, testing and validation over 18 months, market launch in approximately 3 years.

Target Market:

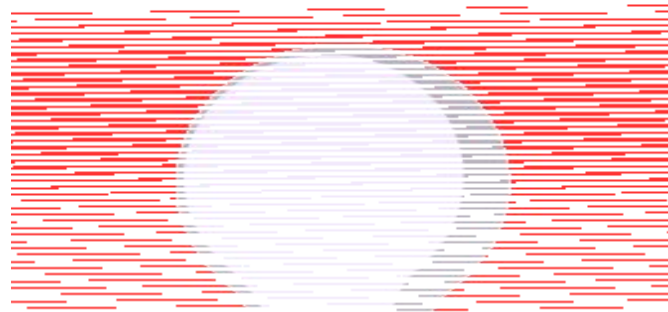
Hospitals will serve as our primary market, supported by insurance reimbursement strategies





Impact And Lessons Learned

- Validated the need for backflow-prevention technology in urology.
- Learned from patient and clinician feedback to guide design.
- Confirmed market viability and clinical relevance.
- Gained technical knowledge in fluid dynamics and CAD modeling.
- Established a roadmap for testing, regulation, and market entry.





THANK YOU

We appreciate your time and consideration.
Please don't hesitate to ask questions or offer
suggestions for improvement.

Any Questions?